Helix VPN — Phase 2 (MVP2) Client Applications

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Comprehensive Technical Specification Package

Version: 1.0
Date: 2026-07-03
Status: Final Draft
Classification: Internal — Development Team

Executive Summary

This package contains the complete technical specification for **Helix VPN Phase 2 (MVP2)** — the client application layer of the Helix VPN ecosystem. MVP2 covers all client-facing applications across **8 platforms** and **7 operating systems**, built on a unified Rust shared core architecture for maximal code reusability (70-85%), minimal footprint, and native performance.

Target Platforms

Platform	OS	UI Framework	Bundle Target	Code Reuse
Desktop	macOS	Tauri v2	< 15 MB	85%
Desktop	Windows	Tauri v2	< 15 MB	80%
Desktop	Linux	Tauri v2	< 15 MB	85%
Mobile	Android	Flutter	< 25 MB	72%
Mobile	iOS	Flutter	< 25 MB	72%
Mobile	HarmonyOS	Flutter (ohos)	< 25 MB	70%

Platform	OS	UI Framework	Bundle Target	Code Reuse
Desktop	Aurora OS	Qt6/QML	< 20 MB	75%
Web	Browser	Extension + Tauri	< 5 MB	45%

Key Metrics

- **Total Documentation:** 24,000+ lines across 10 specification documents
- **Research Base:** 8 wide-exploration research briefs (8,300+ lines)
- **Estimated Implementation:** 36 weeks with 11-person team
- **Code Reuse Target:** 70-85% across all platforms via Rust shared core
- **Protocols Supported:** WireGuard, Shadowsocks, MASQUE/QUIC, OpenVPN

Documentation Index

Core Documents (Read in Order)

#	Document	File	Lines	Description
1	**Overview**	`MVP2_OVERVIEW.md`	696	Executive summary, goals, platform matrix, success criteria
2	**Architecture**	`MVP2_ARCHITECTURE.md`	1,257	System architecture, component diagrams, technology stack decisions
3	**Shared Core**	`MVP2_SHARED_CORE.md`	2,482	Rust core library specification, FFI design, cross-compilation
4	**Desktop Apps**	`MVP2_DESKTOP_APPS.md`	3,225	macOS, Windows,

#	Document	File	Lines	Description
				Linux client specifications with code examples
5	**Mobile Apps**	`MVP2_MOBILE_APPS.md`	4,924	Android, iOS, HarmonyOS client specifications
6	**Web Client**	`MVP2_WEB_CLIENT.md`	2,785	Browser extension, admin panel, PWA specification
7	**Aurora Client**	`MVP2_AURORA_CLIENT.md`	2,564	Aurora OS Qt/QML client with Silica UI
8	**Security & Performance**	`MVP2_SECURITY_PERFORMANCE.md`	2,304	Security architecture, kill switch, CI/CD pipeline
9	**UI/UX Design**	`MVP2_UI_UX_SPEC.md`	1,456	Design system, components, screen specifications
10	**Implementation Roadmap**	`MVP2_IMPLEMENTATION_ROADMAP.md`	2,553	36-week phased plan with milestones

Research Documents

Located in the `research/` directory, containing the raw research that informed all specifications:

Document	Description
`mvp2_wide01.md`	Cross-platform UI frameworks comparison
`mvp2_wide02.md`	OS-specific VPN integration APIs
`mvp2_wide03.md`	VPN protocol implementations in Rust
`mvp2_wide04.md`	Mobile platform implementation details
`mvp2_wide05.md`	Desktop platform implementation details
`mvp2_wide06.md`	Web platform capabilities
`mvp2_wide07.md`	Shared core architecture patterns

Document	Description
`mvp2_wide08.md`	Security, performance, and build pipeline

Architecture Overview

```
                    +------------------------------------+
                    |           helix-core               |
                    |      (Rust Shared Library)         |
                    |  WireGuard | Shadowsocks | MASQUE |
                    |   Crypto   |    TUN      |   DNS  |
                    +---+-----+-----+-----+----+----+
                        |     |      |      |    |
            +-----------+     |      |      |    +-----------+
            |                 |      |      |                |
    +-------v------+ +--------v--+ +--v---------+ +----------
v---------+
    |   Desktop    | |  Mobile   | |  Aurora   | |
Web           |
    |   Tauri v2   | |  Flutter  | |  Qt6/QML  | |  Browser
Extension |
    |              | |           | |           | |  + Tauri
Companion |
    | macOS/Win/Lin| |Android/iOS| | ConnMan   | |  + Admin
Panel     |
    |              | |  HarmonyOS| |  Silica   | |  +
PWA             |
    +--------------+ +-----------+ +-----------+
+--------------------+
```

Quick Start for Stakeholders

For Engineering Leads

Read in order: Overview → Architecture → Shared Core → Implementation Roadmap

For Platform Developers

Read: Overview → Architecture → [Your Platform] → Shared Core → Security & Performance

For UI/UX Designers

Read: Overview → UI/UX Design → [Platform Screens] → Architecture (UI sections)

For Project Managers

Read: Overview → Implementation Roadmap → Architecture (decisions only)

For Security Reviewers

Read: Overview → Security & Performance → Shared Core (security section)

Design Principles

1. **Rust-First Core** — All business logic lives in a memory-safe, high-performance Rust shared library
2. **Maximal Reuse** — 70-85% code reuse across platforms via shared core and unified UI frameworks
3. **Native Performance** — Each platform uses native APIs (NetworkExtension, VpnService, WFP, TUN)
4. **Minimal Footprint** — Desktop < 15MB, Mobile < 25MB, Web < 5MB bundle targets
5. **Security by Design** — Kill switch, leak prevention, post-quantum crypto, secure key storage
6. **Platform Native** — UI follows each platform's design language while maintaining brand consistency

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Implementation Timeline

Phase	Duration	Deliverables
Phase 1: Foundation	Weeks 1-4	Rust core, CI/CD, basic TUN
Phase 2: Protocols	Weeks 3-8	WireGuard, Shadowsocks, DNS
Phase 3: Desktop macOS/Linux	Weeks 6-14	Tauri apps for macOS and Linux
Phase 4: Desktop Windows	Weeks 12-18	Windows Tauri app
Phase 5: Mobile Android	Weeks 14-22	Flutter Android app

Phase	Duration	Deliverables
Phase 6: Mobile iOS	Weeks 20-26	Flutter iOS app
Phase 7: HarmonyOS/ Aurora	Weeks 24-30	Extended platform clients
Phase 8: Web	Weeks 28-34	Browser extension + admin
Phase 9: Launch	Weeks 32-36	Testing, audit, production

Total Expected Duration: 36 weeks (9 months)
Team Size: 11 engineers
Parallel Workstreams: Up to 3 during peak phases

Technology Stack Summary

Layer	Technology	Purpose
Core	Rust + Tokio	VPN engine, protocols, crypto
Desktop UI	Tauri v2 + React/Vue	macOS, Windows, Linux apps
Mobile UI	Flutter + Dart	Android, iOS, HarmonyOS apps
Aurora UI	Qt6 + QML + Silica	Aurora OS app
Web UI	React + TypeScript	Browser extension, admin panel
FFI	UniFFI + flutter_rust_bridge	Language bindings
Protocols	boringtun + shadowsocks-rust	WireGuard + Shadowsocks
Crypto	rustls + x25519-dalek	TLS, key exchange
Build	GitHub Actions + cargo-cross	CI/CD, cross-compilation

Contact & Contribution

- **Project Repository:** `git@github.com:HelixDevelopment/helix_vpn.git`
- **Documentation Updates:** Submit PRs against `docs/research/mvp2/` directory
- **Issue Tracking:** Use GitHub Issues with `mvp2` label

This documentation package was generated on 2026-07-03 as part of the Helix VPN MVP2 planning phase. All specifications are final draft and subject to engineering review before implementation begins.